

PARKING SPACE AVAILABILITY DETECTION USING IMAGE SEGMENTATION AND SPATIAL FILTERING TECHNIQUES

MEMBER OF GROUP:

- ABDULLAH AL-FIRDAUS NUZULA (008)
- FIO ULAA OCTRIYANTI (030)
- DIMAS RAFI IZZULHAQ (084)



BACKGROUND

- Difficulty finding empty parking spaces causes:

Traffic
Congestion

Fuel
Waste

Air
Pollution

- Camera-based monitoring can solve this problem automatically
- This project uses:

Classical Digital
Image Processing

Low-resource
Machine Learning

No Deep
Learning

- Used References from Gonzalez & Woods Digital Image Processing

RESEARCH OBJECTIVES

**Detect parking
slot occupancy
automatically**

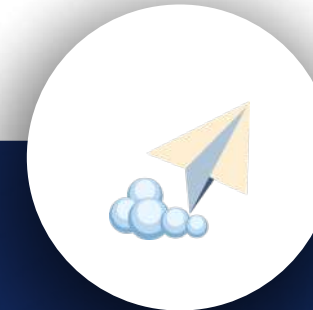


Classify slots into:

- space-empty
- space-occupied



**Evaluate
effectiveness of
classical DIP
methods**



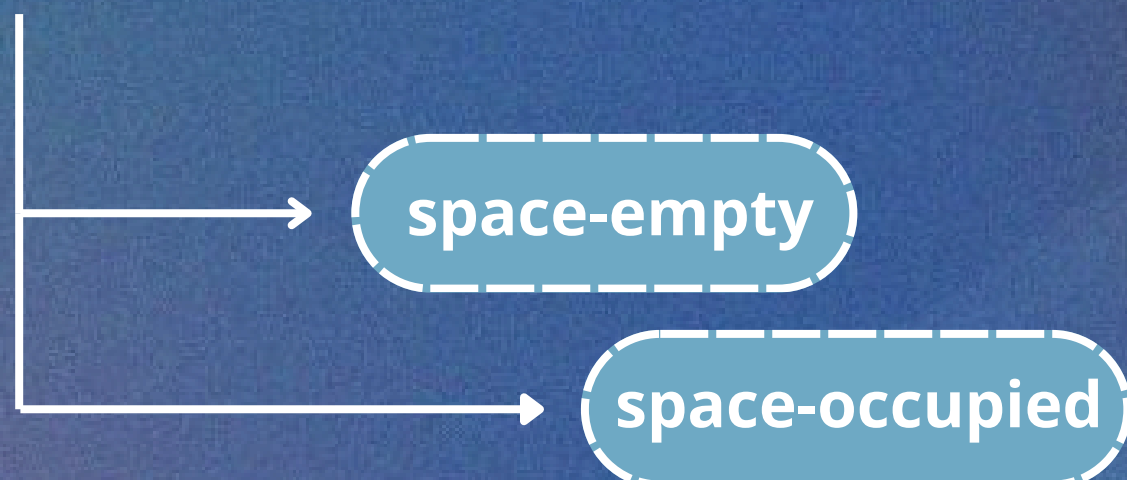
**Build lightweight
parking
monitoring
pipeline**



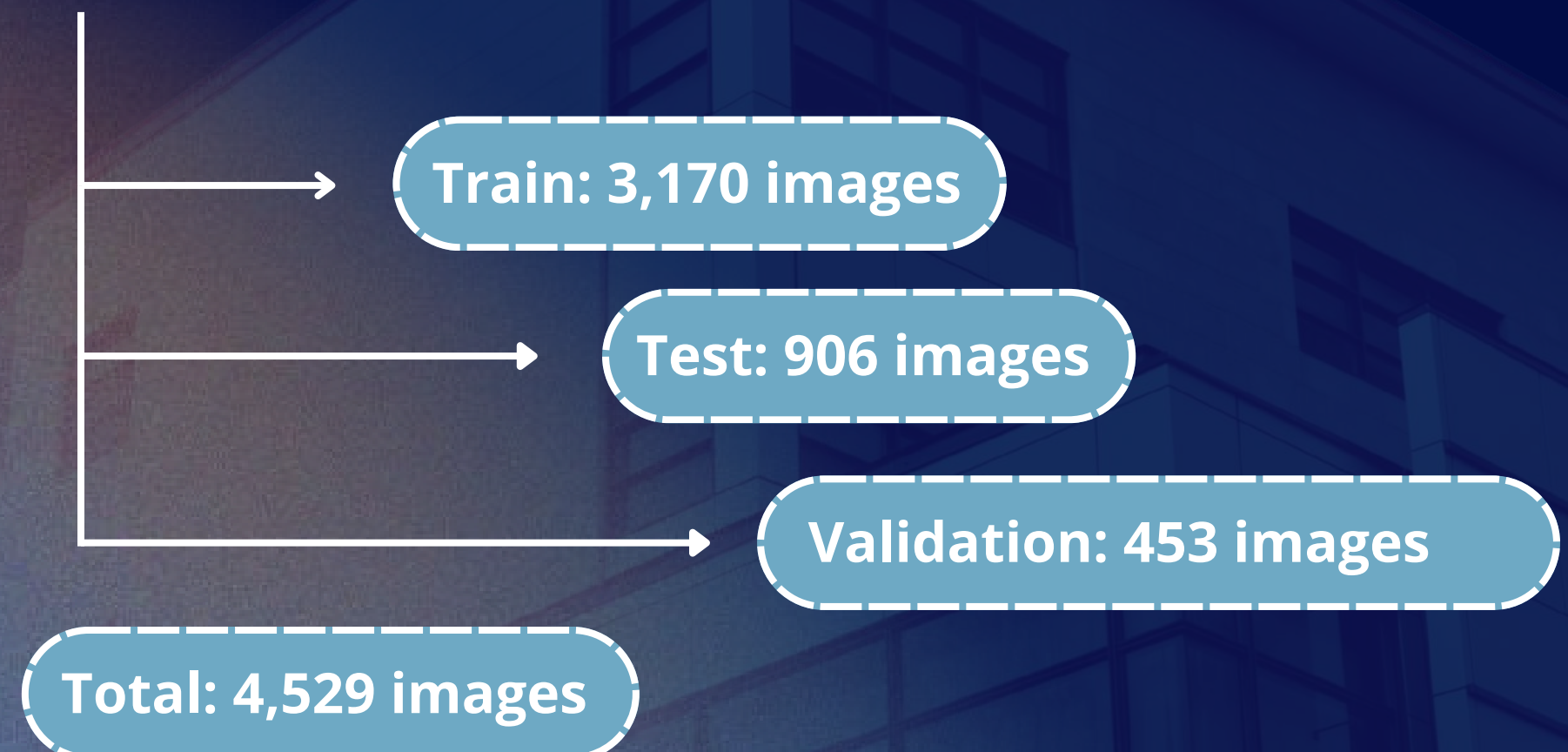
DATASET

Dataset Information:

- Source: Roboflow Dataset
- Format: YOLOv8 Bounding Box
- Resolution: 640×640
- Classes:



Dataset Split (70 : 20 : 10)



DATASET CHALLENGES



Challenge Found

- Lighting variation
- Shadow effects
- Occlusion
- Different vehicle colors
- Inconsistent parking layouts
- Moderate class imbalance

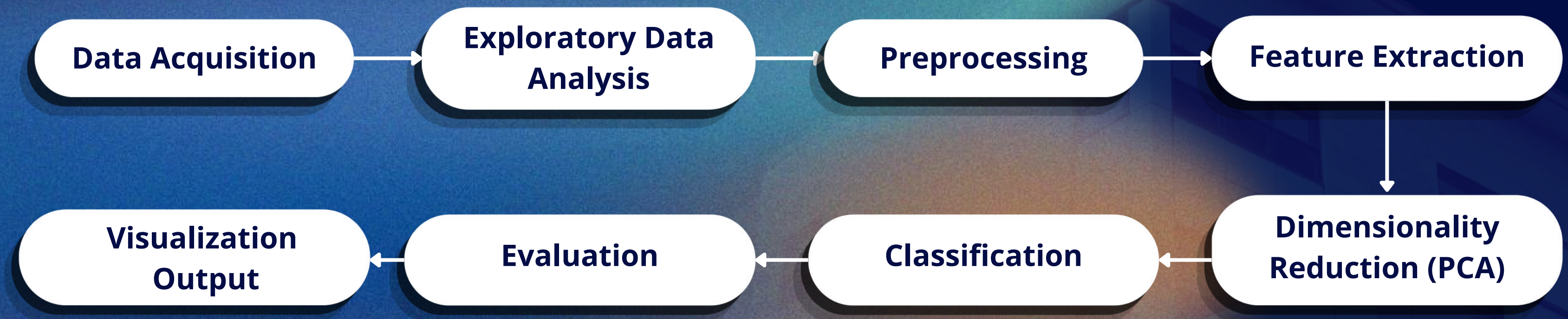
Class Distribution:

Empty: 54.1%

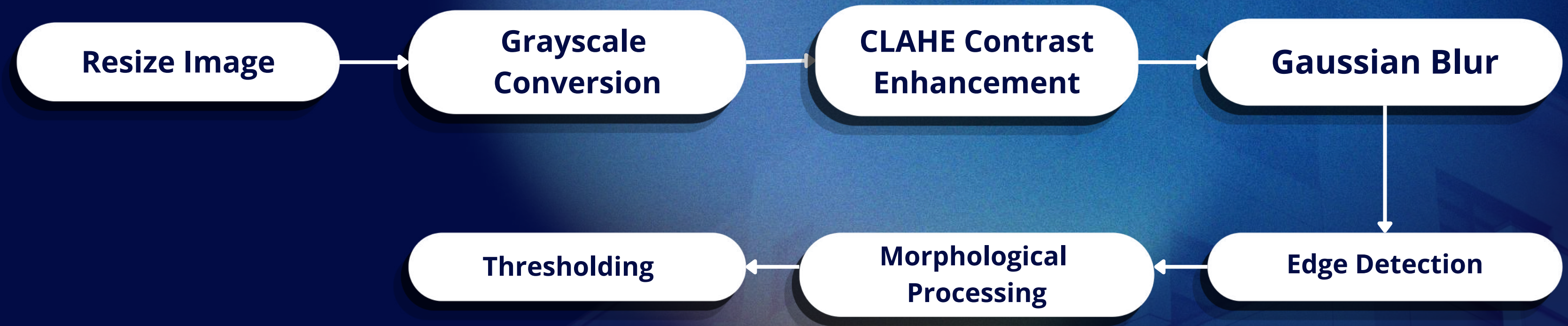
Occupied: 45.9%

**note: per-slot annotation*

RESEARCH PIPELINE



PREPROCESSING



EDGE DETECTION

Compared Method

Sobel

- captures gradient intensity
- sensitive to texture

Laplacian

- detects rapid intensity changes
- high detail response

Canny

- most precise edge detection
- suppresses noise effectively

Final Insight

Canny produced the clearest vehicle boundaries

FEATURE EXTRACTION

Dimensional Feature Vector

Intensity Features

- Pixel Variance
- Mean Intensity
- Standard Deviation

Spatial Features

- Canny Edge Density
- Laplacian Variance
- Sobel Gradient Mean

Histogram Features

- 8-bin grayscale histogram

FEATURE ANALYSIS

Key Findings

Empty Slots

- low variance
- sparse edges
- uniform asphalt texture

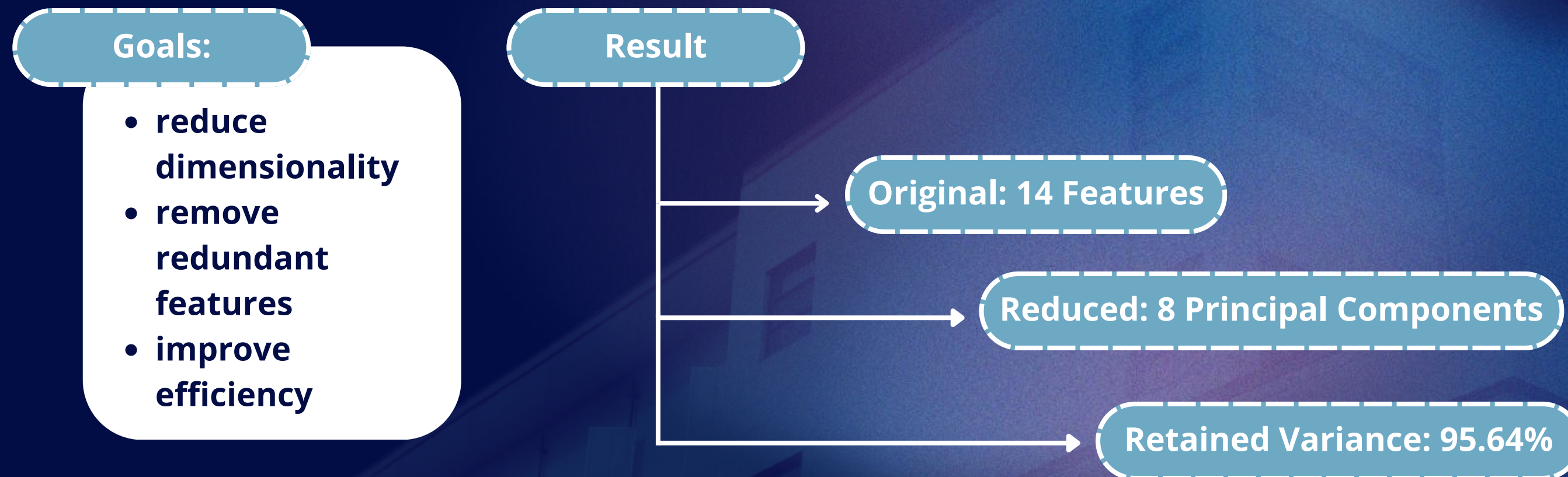
Occupied Slots

- high variance
- dense edge structures
- complex textures

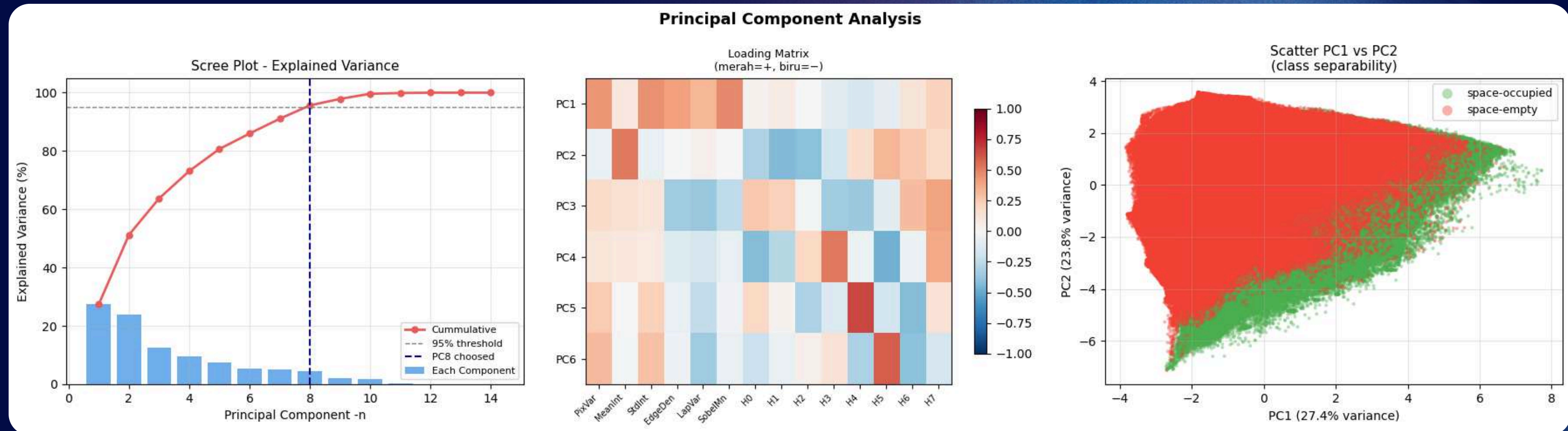
Best Discriminative Features

- Goal:
- reduce dimensionality
- remove redundant features
- improve efficiency

PCA (PRINCIPAL COMPONENT ANALYSIS)



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CLASSIFICATION



Classification Methods

- Random Forest 👍
- Rule-Based Classifier
- SVM Linear
- Gradient Boosting

Original
(14D)

PCA
(8D)

Why Random Forest?

- Handles overlapping features better
- Non-linear decision boundaries
- Stable on heterogeneous parking data

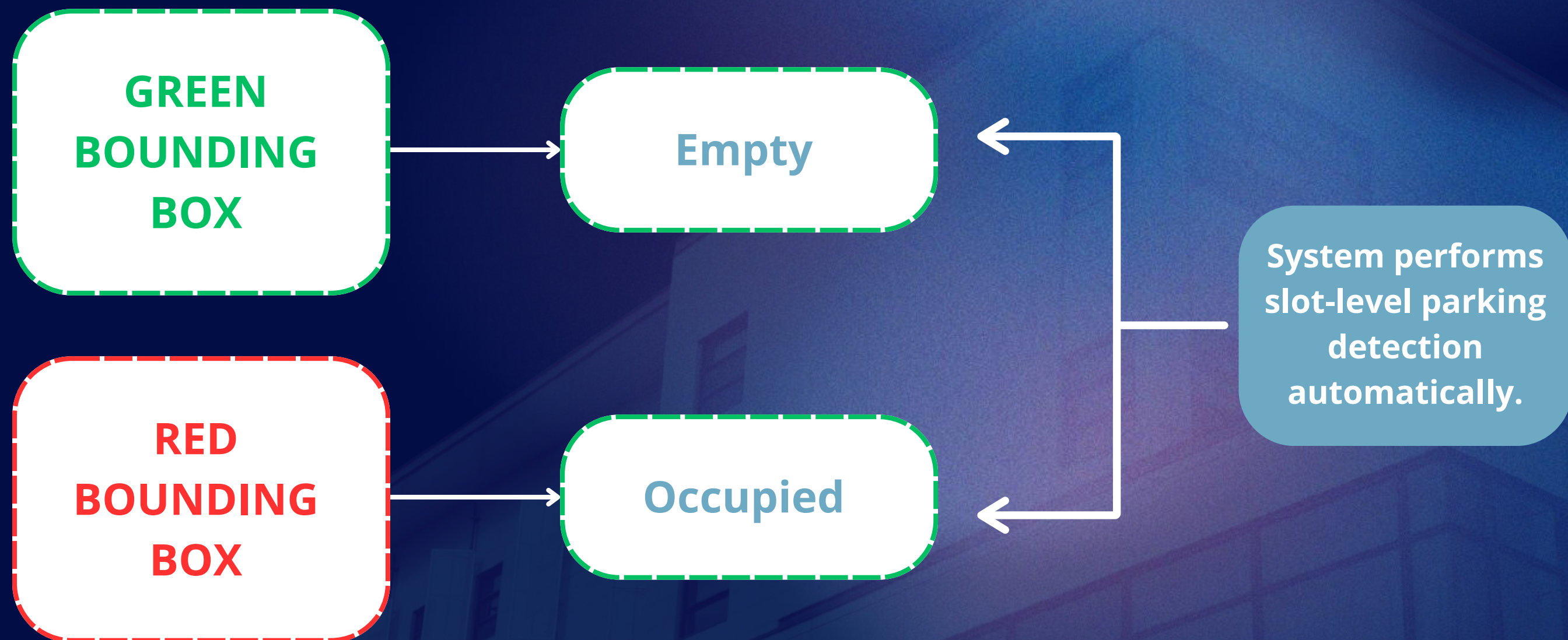
EVALUATION

	Rule Based	Random Forest	SVM	Gradient Boosting
Accuracy	0.7972	0.9241	0.9016	0.9232
F1-Score	0.7957	0.9244	0.9019	0.9235

5 Fold Stratified Cross Validation

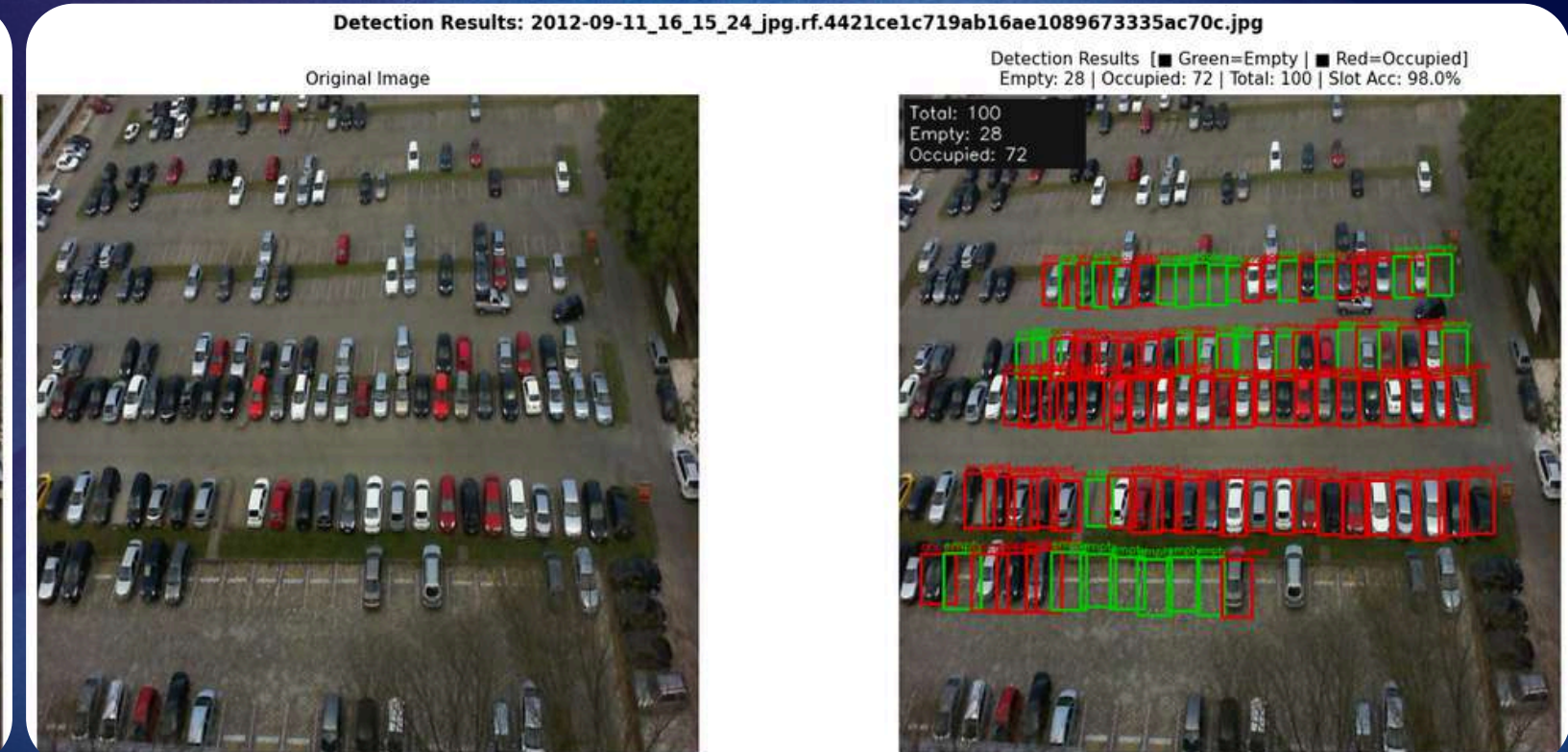
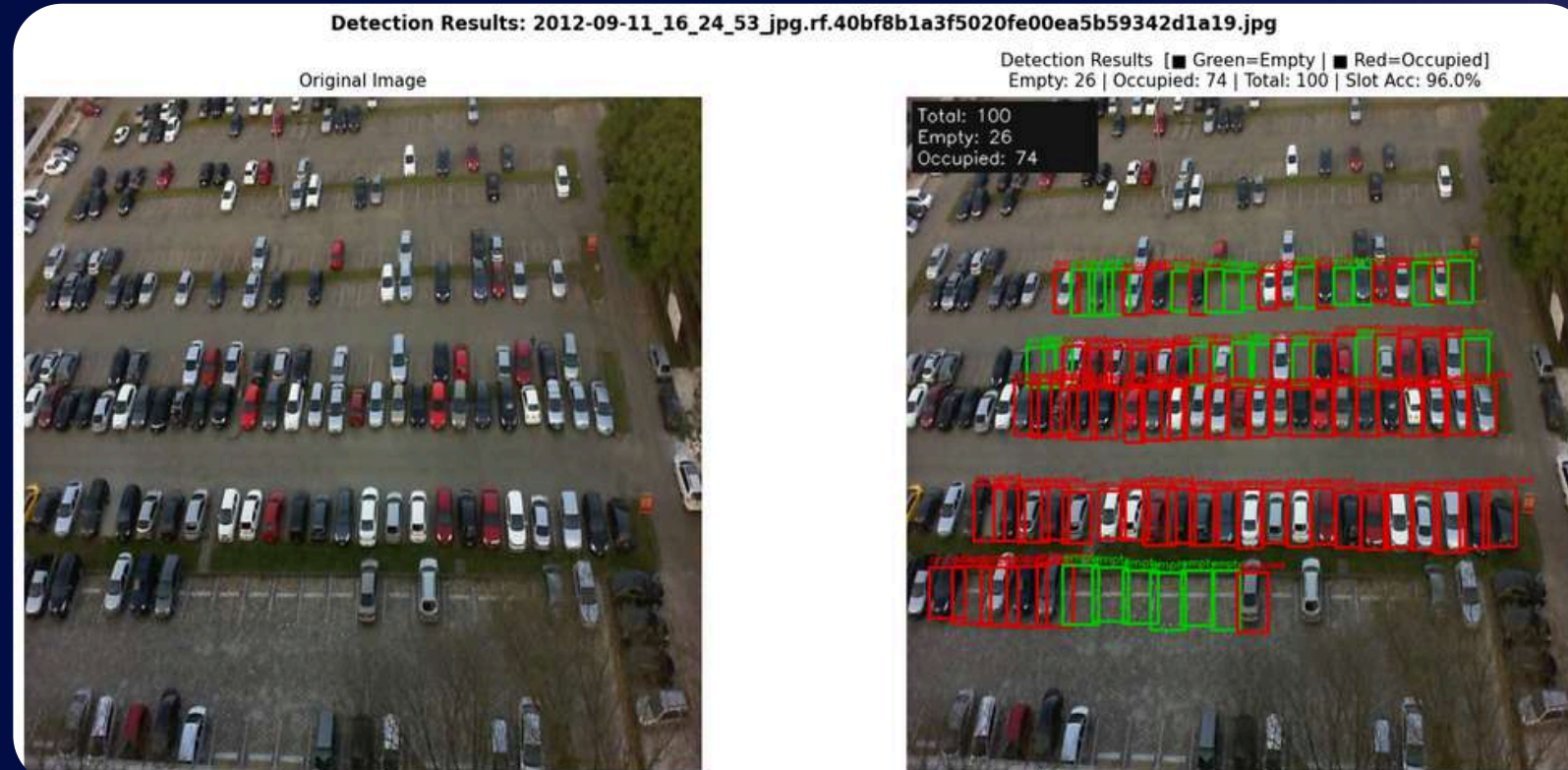
INFERENCE PIPELINE

Final Output



OUTPUT VISUALIZATION

Final Output



CONCLUSION



Conclusion

- Classical DIP methods are effective for parking detection
- Feature extraction successfully separates occupied vs empty slots
- PCA reduces computation while preserving information
- Random Forest provides robust classification performance
- No Deep Learning required



FUTURE DEVELOPMENT

- Real-time CCTV integration
- Web dashboard monitoring
- Smart parking analytics
- Deployment for cafés / urban parking areas

THANK YOU